

St George Girls High School

Year 11 – Higher School Certificate Course

Assessment Task 1

2006



Mathematics

Extension 1

*Time Allowed: 75 Minutes
(plus 5 minutes reading time)*

Instructions to Candidates

1. Write using black or blue pen.
2. Attempt all questions.
3. Start each question on a new page.
4. Show all necessary working.
5. Marks for each question are shown in the right column.
6. Complete cover sheet clearly showing:
 - your name
 - your mathematics class and teacher.

Question 1 – (14 marks) – Start a New Page

Marks

a) Differentiate with respect to x

(i) $y = \log_e(7x+6)$

1

(ii) $y = e^{x^2+2}$

1

(iii) $y = (e^{2x} + 3)^5$

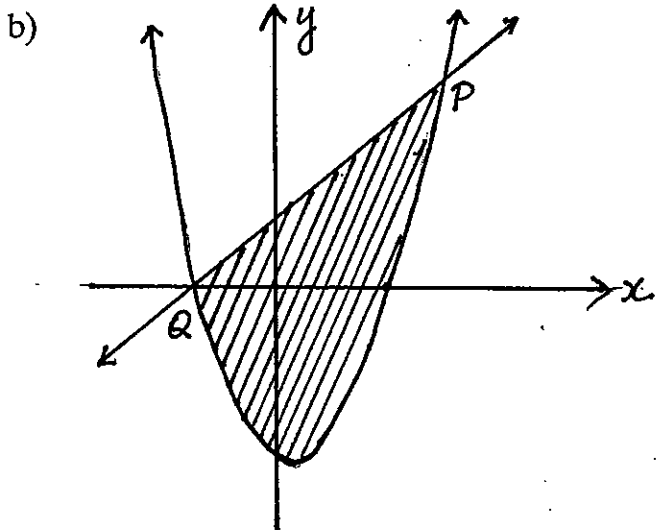
2

(iv) $y = x^3 e^{2x}$

2

(v) $y = \frac{\log_e x}{x}$

2



The graphs of $y = x^2 - x - 6$ and $y = x + 2$ are shown on the diagram.

(i) Find the coordinates of P and Q .

3

(ii) Find the shaded area.

3

Question 2 – (14 marks) – Start a new page

Marks

a) If $\log_a 2 = 1.31$ and $\log_a 3 = 2.07$ find the value of:

(i) $\log_a 6$

1

(ii) $\log_a 4.5$

1

(iii) $\log_a \frac{8}{a^2}$

2

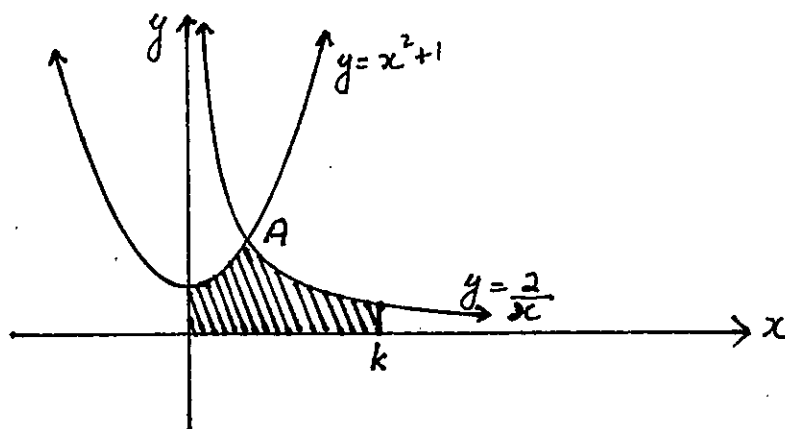
b) Solve the equation $3^x = 17$, giving your answer correct to 3 decimal places.

2

c) Find the equation of the tangent to the curve $y = \log_e(3x-2)+4$ at the point where $x=1$

3

d)



The curves $y = x^2 + 1$ and $y = \frac{2}{x}$ meet at the point A, as shown on the diagram.

(i) Show that A has coordinates (1, 2)

1

(ii) Find the value of k , given that the shaded area is $\frac{10}{3}$ units²

4

Question 3 – (14 marks) – Start a new page

Marks

a) Find the following:

(i) $\int x\sqrt{x} \, dx$

1

(ii) $\int \frac{1}{3x^2} \, dx$

1

(iii) $\int x e^{x^2+1} \, dx$

1

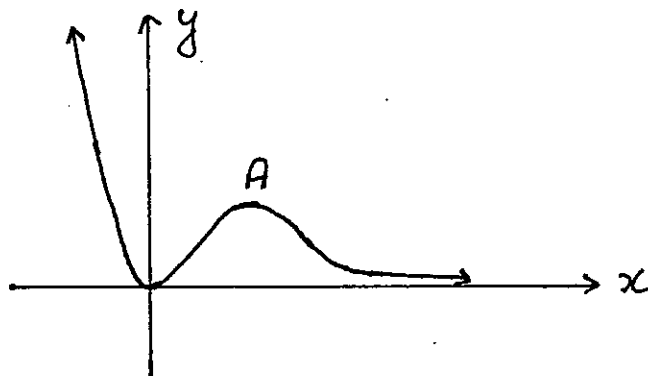
b) Show that $\int_{-6}^2 \frac{1}{x} \, dx = -\log_e 3$

3

c) Use Simpson's Rule with 3 function values to find an approximate value of $\int_0^2 \frac{1}{\sqrt{4+x^2}} \, dx$ correct to 3 decimal places.

3

d)



$y = x^2 e^{-x}$ has a maximum turning point at A, as shown on the graph.

(i) Find the coordinates of A.

4

(ii) The equation $x^2 e^{-x} - k = 0$ has 3 real, distinct roots. Using the graph, or otherwise, write down the possible values of k .

1

Question 4 – (14 marks) – Start a new page

Marks

a) Solve $\log_3(x+3) + \log_3(x-5) = 2$

4

b) The curve $y = f(x)$ has a stationary point at the point $(2, 2)$.

If $\frac{d^2y}{dx^2} = \frac{2}{x^2}$ find $f(x)$

4

c) The section of the curve $y = \frac{x}{\sqrt{x^3+1}}$ from $x=1$ to $x=3$ is rotated about the x axis. Find the volume of the solid of revolution.

3

d) (i) Show that $f(x) = \frac{x^3}{x^2+1}$ is an odd function.

2

(ii) Hence, or otherwise, write down the value of $\int_{-2}^2 \frac{x^3}{x^2+1} dx$

1

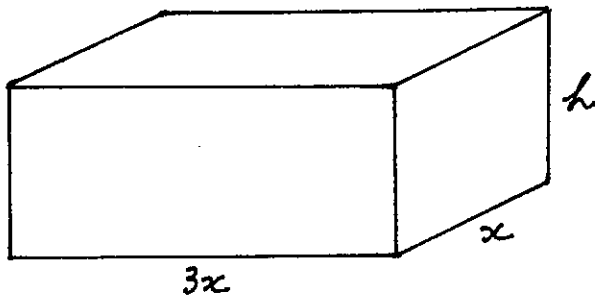
Question 5 – (14 marks) – Start a new page

Marks

a) Evaluate $\int_0^2 (3-2x)^3 dx$

2

b)



A rectangular prism has dimension x , $3x$ and h metres.

(i) If the surface area is 72m^2 show that $h = \frac{36-3x^2}{4x}$

2

(ii) Hence show that the maximum volume of this prism is 36m^3

5

c) (i) Differentiate $y = (x+2)\sqrt{2x+1}$. Express your answer as a single fraction.

2

(ii) Hence or otherwise find $\int_4^{12} \frac{x+1}{\sqrt{2x+1}} dx$

3

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Extension 1 Solutions

HSC Assessment Task I

Question 1

a.) (i) $y = \log_e(7x+6)$

$$\frac{dy}{dx} = \frac{7}{7x+6}$$

(ii) $y = e^{x^2+2}$

$$\frac{dy}{dx} = 2x \cdot e^{x^2+2}$$

(iii) $y = (e^{2x} + 3)^5$

$$\frac{dy}{dx} = 5 \cdot (e^{2x} + 3)^4 \cdot 2e^{2x}$$
$$= 10e^{2x} (e^{2x} + 3)^4$$

(iv) $y = x^3 e^{2x}$

$$\frac{dy}{dx} = e^{2x} \cdot (3x^2) + x^3 (2e^{2x})$$
$$= x^2 e^{2x} (3 + 2x)$$

(v) $y = \frac{\log_e x}{x}$

$$\frac{dy}{dx} = x \cdot \left(\frac{1}{x} \right)' - \frac{\log_e x \cdot (1)}{x^2}$$
$$= \frac{1 - \log_e x}{x^2}$$

b.) (i) $x^2 - x - 6 = x + 2$

$$x^2 - 2x - 8 = 0$$

$$(x+2)(x-4) = 0$$

$$x = -2, 4$$

when $x = -2$, $y = 0$

when $x = 4$, $y = 6$

$\therefore$ P is $(4, 6)$ and Q is $(-2, 0)$

(ii) $A = \int_{-2}^4 x+2-(x^2-x-6) dx$

$$= \int_{-2}^4 (2x+8-x^2) dx$$

$$= \left[x^2 + 8x - \frac{x^3}{3} \right]_{-2}^4$$

$$= (16 + 32 - \frac{64}{3}) - (4 - 16 + \frac{8}{3})$$

$$= 36 \text{ units}^2$$

Question 2

a.) $\log_a 2 = 1.31$ $\log_a 3 = 2.07$

(i) $\log_a 6 = \log_a (2 \times 3)$

$$= \log_a 2 + \log_a 3$$
$$= 1.31 + 2.07$$
$$= 3.38$$

(ii) $\log_a 4.5 = \log_a \left(\frac{3^2}{2} \right)$

$$= 2\log_a 3 - \log_a 2$$
$$= 2 \times 2.07 - 1.31$$
$$= 2.83$$

(iii) $\log_a \frac{8}{a^2} = \log_a 2^3 - \log_a a$

$$= 3\log_a 2 - 2$$
$$= 3 \times 1.31 - 2$$
$$= 1.93$$

b.) $3^x = 17$

$$\log_e 3^x = \log_e 17$$

$$x = \frac{\log_e 17}{\log_e 3}$$

$$= 2.579 \text{ (3 d.p.)}$$

$$c.) y = \log_e(3x-2) + 4 \quad x=1$$

$$\frac{dy}{dx} = \frac{3}{3x-2}$$

$$\text{when } x=1, \frac{dy}{dx} = \frac{3}{1}$$

$$\text{when } x=1, y = \log_e 1 + 4 = 4$$

$$m=1 \quad (1, 4)$$

$$y-4 = 1(x-1)$$

$\therefore$ equation of the tangent is $y = x + 3$

$$d.) i) \text{ Sub. } (1, 2) \text{ in } y = x^2 + 1$$

$$\text{RHS} = 2$$

$$= \text{LHS}$$

$$\text{Sub. } (1, 2) \text{ in } y = \frac{2}{x}$$

$$\text{RHS} = 2$$

$$= \text{LHS}$$

$\therefore$ A has coordinates (1, 2) since it satisfies both equations.

$$ii) \frac{16}{3} = \int_0^1 x^2 + 1 dx + \int_1^k \frac{2}{x} dx$$

$$= \left[\frac{x^3}{3} + x \right]_0^1 + \left[2 \log_e x \right]_1^k$$

$$= \left(\frac{1}{3} + 1 - 0 \right) + (2 \log_e k - 2 \log_e 1)$$

$$= \frac{4}{3} + 2 \log_e k$$

$$2 \log_e k = 2$$

$$\log_e k = 1$$

$$\therefore k = e$$

Question 3

$$a.) i) \int x \sqrt{x} dx = \int x^{\frac{3}{2}} dx$$

$$= \frac{2x^{\frac{5}{2}}}{\frac{5}{2}} + C$$

$$ii) \int \frac{1}{3x^2} dx = \int \frac{1}{3} x^{-2} dx$$

$$= \frac{1}{3} \cdot x^{-1} + C$$

$$= \frac{-1}{3x} + C$$

$$iii) \int x \cdot e^{x^2+1} dx = \frac{1}{2} e^{x^2+1} + C$$

$$b) \int_{-6}^{-2} \frac{1}{x} dx = \left[\log_e x \right]_{-6}^{-2}$$

$$= (\log_e -2 - \log_e -6)$$

$$= \log_e \left(\frac{-2}{-6} \right)$$

$$= \log_e \left(\frac{1}{3} \right)$$

$$= \log_e 3^{-1}$$

$$= -\log_e 3$$

$$c.) \int_0^2 \frac{1}{\sqrt{4+x^2}} dx$$

x	0	1	2
$f(x)$	0.5	0.447	0.354

$$\int_0^2 \frac{1}{\sqrt{4+x^2}} dx \approx \frac{2}{6} [f(0) + 4f(1) + f(2)]$$

$$= \frac{1}{3} [0.5 + 4 \times 0.447 + 0.354]$$

$$= 0.881 \text{ (3 d.p.)}$$

$$d) y = x^2 e^{-x}$$

$$(i) \frac{dy}{dx} = e^{-x} (2x) + x^2 (-e^{-x})$$

$$\frac{dy}{dx} = x e^{-x} (2 - x)$$

stat. pts occur when $\frac{dy}{dx} = 0$

$$x e^{-x} (2 - x) = 0$$

$$x = 0, 2$$

$$\text{when } x = 2, y = \frac{4}{e^2}$$

$\therefore$ A has coordinates $(2, \frac{4}{e^2})$

$$(ii) 0 < x < 2$$

Question 4

$$a.) \log_3 (x+3) + \log_3 (x-5) = 2$$

$$\log_3 ((x+3)(x-5)) = 2$$

$$(x+3)(x-5) = 3^2$$

$$x^2 - 2x - 15 = 9$$

$$x^2 - 2x - 24 = 0$$

$$(x+4)(x-6) = 0$$

$$x = -4, 6$$

$\therefore x = 6$, since $x+3 > 0$
and $x-5 > 0$

$$b) \frac{d^2y}{dx^2} = \frac{2}{x^2} \quad \text{stat. pt. @ } (2, 2)$$

$$\frac{dy}{dx} = \int 2x^{-2} dx$$

$$= -2x^{-1} + C_1$$

when $\frac{dy}{dx} = 0$, $x = 2$

$$\frac{dy}{dx} = -\frac{2}{x} + C_1$$

$$0 = -\frac{2}{2} + C_1$$

$$\therefore C_1 = 1$$

$$\frac{dy}{dx} = -\frac{2}{x} + 1$$

$$y = \int \left(-\frac{2}{x} + 1\right) dx$$

$$= -2 \log x + x + C_2$$

when $x = 2$, $y = 2$

$$2 = -2 \log 2 + 2 + C_2$$

$$\therefore C_2 = \log 4$$

$$\therefore f(x) = x - 2 \log x + \log 4$$

$$c) y = \frac{x}{\sqrt{x^3+1}} \quad x=1 \text{ to } x=3$$

$$y^2 = \frac{x^2}{x^3+1}$$

$$V = \pi \int_1^3 \frac{x^2}{x^3+1} dx$$

$$= \frac{1}{3} \pi [\log(x^3+1)]_1^3$$

$$= \frac{\pi}{3} (\log 28 - \log 2)$$

$$= \frac{\pi \log 14}{3} \text{ units}^3$$

$$d.) (i) f(x) = \frac{x^3}{x^2+1}$$

$$f(-x) = \frac{-x^3}{x^2+1}$$

$$-f(x) = -\left(\frac{x^3}{x^2+1}\right)$$

$$= f(-x)$$

$\therefore$ odd function

$$(ii) \int_{-2}^2 \frac{x^3}{x^2+1} dx = 0$$

Question 5

$$\begin{aligned} a.) \int_0^2 (3-2x)^3 dx &= \left[\frac{(3-2x)^4}{4 \times -2} \right]_0^2 \\ &= \left[\frac{(3-2x)^4}{-8} \right]_0^2 \\ &= -\frac{1}{8} + \frac{81}{8} \\ &= 10 \end{aligned}$$

$$\begin{aligned} b.) i) A &= 2(3x \times x) + 2(3x \times h) + 2(x \times h) \\ &= 6x^2 + 6xh + 2xh \\ &= 6x^2 + 8xh \\ 72 &= 6x^2 + 8xh \\ 8xh &= 72 - 6x^2 \\ h &= \frac{72 - 6x^2}{8x} \\ \therefore h &= \frac{36 - 3x^2}{4x} \end{aligned}$$

$$\begin{aligned} (ii) V &= 3x^2h \\ &= 3x^2 \cdot \left(\frac{36 - 3x^2}{4x} \right) \\ &= 27x - \frac{9x^3}{4} \end{aligned}$$

$$\frac{dV}{dx} = 27 - \frac{27x^2}{4}$$

$$\begin{aligned} \text{stat. pt occurs when } \frac{dV}{dx} &= 0 \\ 27 &= \frac{27x^2}{4} \end{aligned}$$

$$x^2 = 4$$

$$\therefore x = 2, x > 0$$

$$\frac{d^2V}{dx^2} = -\frac{27x}{2}$$

$$\text{when } x=2, \frac{d^2V}{dx^2} = -27 < 0$$

$$\begin{aligned} \therefore \text{max. occurs when } x &= 2 \\ \text{when } x=2, V &= 27 \times 2 - \frac{1 \times 6}{4} \\ &= 36 \text{ m}^3 \end{aligned}$$

$$\begin{aligned} c.) (i) y &= (x+2)\sqrt{2x+1} \\ &= (x+2)(2x+1)^{\frac{1}{2}} \\ \frac{dy}{dx} &= (2x+1)^{-\frac{1}{2}} \cdot 1 + (x+2) \cdot \frac{1}{2}(2x+1)^{-\frac{1}{2}} \\ &= \sqrt{2x+1} + \frac{(x+2)}{\sqrt{2x+1}} \\ &= \frac{2x+1 + x+2}{\sqrt{2x+1}} \\ &= \frac{3x+3}{\sqrt{2x+1}} \end{aligned}$$

$$\begin{aligned} (ii) \int_4^{12} \frac{3x+3}{\sqrt{2x+1}} dx &= (x+2)\sqrt{2x+1} \\ \therefore \frac{1}{3} \int_4^{12} \frac{3(x+1)}{\sqrt{2x+1}} dx &= \int_4^{12} \frac{x+1}{\sqrt{2x+1}} dx \\ &= \left[\frac{1}{3} (x+2)\sqrt{2x+1} \right]_4^{12} \end{aligned}$$

$$= \frac{1}{3} (14 \times 5 - 6 \times 3)$$

$$= 17 \frac{1}{3}$$